

Barriers and Challenges of Computing Students in an Online Learning Environment: Insights from One Private University in the Philippines

- A real-Philippines study detailing challenges students faced in online learning during lockdown — good for your background on digital divide, internet access, readiness issues.

Innovation in Education: Developing and Assessing Gamification in the University of the Philippines Open University Massive Open Online Courses

- A recent, local study about gamified online learning (badges, leaderboards, etc.), giving evidence about how gamification affects motivation and completion rates useful for your analysis/solution part.

AI-powered Digital Framework for Personalized Economical Quality Learning at Scale

- A recent paper proposing an AI-powered e-learning framework focusing on quality, scalability, and equity — relevant to your AI + web/mobile online classroom with educational equity.

Could AI Democratise Education? Socio-Technical Imaginaries of an EdTech Revolution

- A critical discussion about the promises and risks of AI in education — useful to support your ethical or social concerns analysis.

Fabito, B. S., Trillanes, A. O., & Sarmiento, J. R. (2021). Barriers and challenges of computing students in an online learning environment: Insights from one private university in the Philippines. *International Journal of Computing Sciences Research*, 5(1), 441–458. stepacademic.net+2ResearchGate+2

Moldez, C., Crisanto, M. A., Cerdeña, M. G. R., Maranan, D. S., & Figueroa, R. (2024). Innovation in education: Developing and assessing gamification in the University of the Philippines Open University massive open online courses. *arXiv preprint arXiv:2409.03309*. [arXiv](https://arxiv.org/abs/2409.03309)

VatandoustMohammadieh, M., Mohajeri, M. M., Keramati, A., & Nili Ahmadabadi, M. (2024). **AI-powered digital framework for personalized economical quality learning at**

scale. *arXiv preprint arXiv:2412.04483*. [arXiv+1](#)

Bulathwela, S., Pérez-Ortiz, M., Holloway, C., & Shawe-Taylor, J. (2021). Could AI democratise education? Socio-technical imaginaries of an EdTech revolution. *arXiv preprint arXiv:2112.02034*.

Republic Act No. 10173 – Data Privacy Act of 2012	
Definitions, scope, and general provisions (e.g. what “personal information,” “data subject,” “consent,” “processing,” etc. mean)	Chapter I – General Provisions, especially Section 3: Definition of Terms
Legal basis for processing personal information; rules about lawful processing of personal information (principles, consent, legitimate purpose, limitations)	Chapter III – Processing of Personal Information → includes Section 11 (General Data Privacy Principles); Section 12 (Criteria for Lawful Processing); Section 13 (Sensitive Personal Information & Privileged Information)
Rights of the data subject (right to be informed, access, correction or deletion, data portability, withdraw consent, etc.)	Chapter IV – Rights of the Data Subject, notably Section 16 (Rights of the Data Subject), Section 18 (Right to Data Portability), plus related subsections.
Security obligations (protection of personal information, safeguards, security measures for storage and processing)	Chapter V – Security of Personal Information, especially Section 20 (Security of Personal Information)
Accountability of controllers/processors — responsibility for compliance, liability if misuse or breach occurs	Chapter VI – Accountability for Transfer of Personal Information, Section 21 (Principle of Accountability)
Penalties for violations (unauthorized processing, disclosure, improper disposal, breach, etc.) — useful if you discuss consequences for non-compliance	Chapter VIII – Penalties, various sections (e.g. Section 25: Unauthorized Processing, Section 32: Unauthorized Disclosure, etc.)

Republic Act No. 10173, Data Privacy Act of 2012, § 3, 11, 12, 13, 16, 18, 20, 21, 25, 32 (2012). National Privacy Commission.

https://privacy.gov.ph/data-privacy-act/?utm_source=chatgpt.com